



1. Determine which transfer functions are stable:

a) $H(s) = \frac{s-1}{(s+2)^2}$.

b) $H(s) = \frac{1}{(s^2+4)}$.

c) $H(s) = \frac{s}{(s+3)^2}$.

d) $H(s) = \frac{s}{(s^2-4)}$

e) $H(s) = \frac{1}{s(s+1)}$

f) $H(s) = \frac{1}{s^2(s+1)}$

For the unstable systems, give an example of a bounded input that yields an unbounded output.

2. For the systems given below, calculate the DC gain. Then, calculate the step responses using partial fraction expansions and compare the steady-state values to the values predicted by the DC gain.

a) $H(s) = \frac{2}{s^2+2s+1}$

b) $H(s) = \frac{-s-2}{s^2+2s+2}$

3. Consider the circuit in Figure 1. All the initial conditions in the circuit are zero.

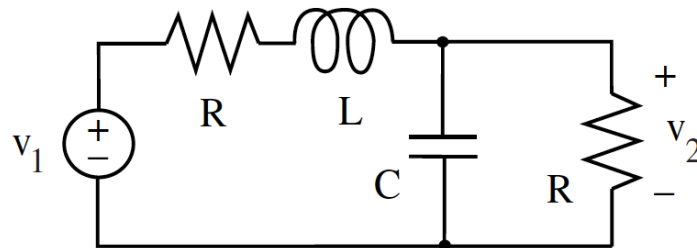


Figure 1: Question-3

- a) Calculate the transfer function from v_1 to v_2 . Show that $H(s) = \frac{1}{(s^2+2s+2)}$ when $R = L = C = 1$.
- b) Using partial fraction expansions, calculate the response $v_2(t)$ to $v_1(t) = 5\cos(t)$. Indicate which part of the response is the transient response and which part is the steady-state response.

- c) Is there a frequency ω such that the steady-state response to $\cos(\omega t)$ is zero for all t ? Is there a frequency ω such that the steady-state response to $\cos(\omega t)$ is proportional to $\sin(\omega t)$? Give reasons for both the scenarios.

For parts (b) and (c), take $R = L = C = 1$.

4. Consider the system:

$$H(s) = \frac{1}{s(s+1)}$$

- Find the response $y(t)$ for input $x(t) = 1$.
- Find the response $y(t)$ for input $x(t) = \sin t$.
- Is a system BIBO stable if its response to a step input is bounded?
- Is a system BIBO unstable if its response to a step input is unbounded?
- Is the response of $H(s) = \frac{1}{(s+1)^2(s+4)}$ to $X(s) = \frac{s+3}{s(s+4)}$ bounded? Does it converge to a steady-state value? If so, to what value?
- Is the response of $H(s) = \frac{1}{(s+1)^2(s+4)}$ to $X(s) = \frac{1}{(s^2+1)^2}$ bounded? Does it converge to a steady-state value? If so, to what value?

For parts (e) and (f), solve without doing partial fraction expansion. Also, give reasons for your answers.

5. Find the transfer function $W(s)/R(s)$ for the following system: Given that the steady-

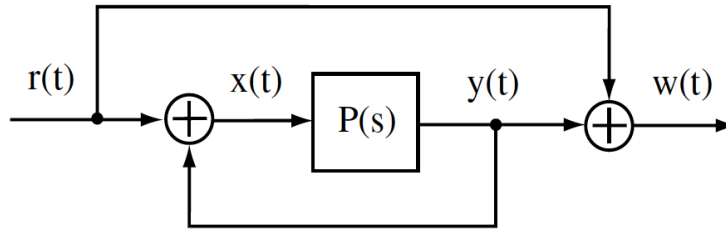


Figure 2: Question-3

state response of $P(s)$ is $y_{ss}(t) = \sin(10t - 60^\circ)$ when $x(t) = \sin(10t)$, what is the steady-state response $w_{ss}(t)$ of the above system when $r(t) = \sin(10t)$? Give the condition that $P(s)$ must satisfy for the result to be true.

6. For the system $P(s) = \frac{s}{(s+1)^2}$:

- Calculate the transient response $y_{tr}(t)$ for the input $x(t) = 2\cos(t)$.
- Without performing a partial fraction expansion, give the value of the steady state output $y_{ss}(t)$ of the system of part (a) for a constant input $x(t) = 2$. If the steady-state output does not exist, explain why.
- System with transfer function $P_1(s)$ has steady-state response $2\cos(t - 30^\circ)$ when an input $x(t) = \cos(t)$ is applied to the system. Another system with transfer function $P_2(s)$ has steady-state response $3\sin(t + 60^\circ)$ when an input $x(t) = \sin(t)$ is applied. What is the steady-state response of the cascade system $P(s) = P_1(s)P_2(s)$ if an input $x(t) = \cos(t)$ is applied to the system?